# 4 Formal Logic Cheatsheet

## 4.1 Propositional Logic

* **Proposition** is a statement which can be either true or false. Truth-bearer
* **Alphabet** of propositional logic consists of (1) atomic symbols and (2) operator symbols.
* **Atomic formula** (**atom**) is an irreducible formula without logical connectives.
  - Propositional **variables**: $A, B, C, \ldots, Z$. With indices, if needed: $A_1, A_2, \ldots, Z_1, Z_2, \ldots$.
  - Logical **constants**: $\top$ for always true proposition (*tautology*), $\bot$ for always false proposition (*contradiction*).
* **Logical connectives** (**operators**):

| Type | Natural meaning | Symbolization |
|---|---|---|
| Negation | It is not the case that $\mathcal{P}$.<br>It is false that $\mathcal{P}$.<br>It is not true that $\mathcal{P}$. | $\neg\mathcal{P}$ |
| Conjunction | Both $\mathcal{P}$ and $\mathcal{Q}$.<br>$\mathcal{P}$ but $\mathcal{Q}$.<br>$\mathcal{P}$, although $\mathcal{Q}$. | $\mathcal{P} \wedge \mathcal{Q}$ |
| Disjunction | Either $\mathcal{P}$ or $\mathcal{Q}$ (or both).<br>$\mathcal{P}$ unless $\mathcal{Q}$. | $\mathcal{P} \vee \mathcal{Q}$ |
| Exclusive or (Xor) | Either $\mathcal{P}$ or $\mathcal{Q}$ (but not both).<br>$\mathcal{P}$ xor $\mathcal{Q}$. | $\mathcal{P} \oplus \mathcal{Q}$ |
| Implication (Conditional) | If $\mathcal{P}$, then $\mathcal{Q}$.<br>$\mathcal{P}$ only if $\mathcal{Q}$.<br>$\mathcal{Q}$ if $\mathcal{P}$. | $\mathcal{P} \to \mathcal{Q}$ |
| Biconditional | $\mathcal{P}$, if and only if $\mathcal{Q}$.<br>$\mathcal{P}$ iff $\mathcal{Q}$.<br>$\mathcal{P}$ just in case $\mathcal{Q}$. | $\mathcal{P} \leftrightarrow \mathcal{Q}$ |

* **Sentence** of propositional logic is defined inductively: Well-formed formula (WFF)
  1. Every propositional variable/constant is a sentence.
  2. If $\mathcal{A}$ is a sentence, then $\neg\mathcal{A}$ is a sentence.
  3. If $\mathcal{A}$ and $\mathcal{B}$ are sentences, then $(\mathcal{A} \wedge \mathcal{B})$, $(\mathcal{A} \vee \mathcal{B})$, $(\mathcal{A} \to \mathcal{B})$, $(\mathcal{A} \leftrightarrow \mathcal{B})$ are sentences.
  4. Nothing else is a sentence.
* Well-formed formulae grammar: Backus-Naur form (BNF)

$$
\begin{aligned}
\langle \textit{sentence} \rangle &::= \langle \textit{constant} \rangle \\
&\quad\;\mid \langle \textit{variable} \rangle \\
&\quad\;\mid \neg\, \langle \textit{sentence} \rangle \\
&\quad\;\mid \text{`(' } \langle \textit{sentence} \rangle\ \langle \textit{binop} \rangle\ \langle \textit{sentence} \rangle \text{ `)'} \\
\langle \textit{constant} \rangle &::= \top \mid \bot \\
\langle \textit{variable} \rangle &::= A \mid \ldots \mid Z \mid A_1 \mid \ldots \mid Z_n \\
\langle \textit{binop} \rangle &::= \wedge \mid \vee \mid \oplus \mid \to \mid \leftarrow \mid \leftrightarrow
\end{aligned}
$$

* **Literal** is a propositional variable or its negation: $\mathcal{L}_i = X_i$ (*positive* literal), $\mathcal{L}_j = \neg X_j$ (*negative* literal).
* **Argument** is a set of logical statements, called *premises*, intended to support or infer a claim (*conclusion*):

$$\underbrace{\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n}_{\textit{premises}} \quad \therefore \quad \underbrace{C}_{\textit{conclusion}}$$

"therefore"

* An argument is **valid** if whenever all the premises are true, the conclusion is also true. Validity
* An argument is **invalid** if there is a case (*a counterexample*) when all the premises are true, but the conclusion is false.

## 4.2 Semantics of Propositional Logic

* **Valuation** is any assignment of truth values to propositional variables. Interpretation
* $\mathcal{A}$ is a **tautology** (**valid**) iff it is true on *every* valuation. Might be symbolized as "$\vDash \mathcal{A}$".
* $\mathcal{A}$ is a **contradiction** iff it is false on *every* valuation. Might be symbolized as "$\mathcal{A} \vDash$ ".
* $\mathcal{A}$ is a **contingency** iff it is true on some valuation and false on another. In other words, a **contingent** proposition is neither a tautology nor a contradiction.
* $\mathcal{A}$ is **satisfiable** iff it is true on *some* valuation. Satisfiability
* $\mathcal{A}$ is **falsifiable** iff it is not valid, *i.e.* it is false on *some* valuation. Falsifiability
* $\mathcal{A}$ and $\mathcal{B}$ are **equivalent** (symbolized as $\mathcal{A} \equiv \mathcal{B}$) iff, for every valuation, their truth values agree, *i.e.* there is no valuation in which they have opposite truth values. Equivalence check
* $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ are **consistent** (**jointly satisfiable**) iff there is *some* valuation which makes them all true. Sentences are **inconsistent** (**jointly unsatisfiable**) iff there is *no* valuation that makes them all true. Consistency
* The sentences $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ **entail** the sentence $C$ (symbolized as $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n \vDash C$) if there is no valuation which makes all of $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ true and $C$ false. Semantic entailment
* If $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n \vDash C$, then the argument $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n \therefore C$ is **valid**.

  *Validity check examples*:

| $A$ | $B$ | $A \to B$ | $\boldsymbol{A}$ | valid $\therefore$ | $\boldsymbol{B}$ | $\neg A$ | $\to$ | $\neg B$ | valid $\therefore$ | $B \to A$ | $A \to B$ | $\boldsymbol{B}$ | invalid $\therefore$ | $\neg$ | $(B \to A)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | **1** | **0** | · | **0** | 1 | **1** | 1 | ✓ | **1** | **1** | **0** | · | **0** | 1 |
| 0 | 1 | **1** | **0** | · | **1** | 1 | **0** | 0 | · | **0** | **1** | **1** | ✓ | **1** | 0 |
| 1 | 0 | **0** | **1** | · | **0** | 0 | **1** | 1 | ✓ | **1** | **0** | **0** | · | **0** | 1 |
| 1 | 1 | **1** | **1** | ✓ | **1** | 0 | **1** | 0 | ✓ | **1** | **1** | **1** | ✗ | **0** | 1 |

| $R$ | $S$ | $T$ | $R \vee S$ | $S \vee T$ | $\neg R$ | invalid $\therefore$ | $S \wedge T$ | $(R \wedge S)$ | $\to$ | $T$ | valid $\therefore$ | $R$ | $\to$ | $(S \to T)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | **0** | **0** | **1** | · | **0** | 0 | **1** | 0 | ✓ | 0 | **1** | 1 |
| 0 | 0 | 1 | **0** | **1** | **1** | · | **0** | 0 | **1** | 1 | ✓ | 0 | **1** | 1 |
| 0 | 1 | 0 | **1** | **1** | **1** | ✗ | **0** | 0 | **1** | 0 | ✓ | 0 | **1** | 0 |
| 0 | 1 | 1 | **1** | **1** | **1** | ✓ | **1** | 0 | **1** | 1 | ✓ | 0 | **1** | 1 |
| 1 | 0 | 0 | **1** | **0** | **0** | · | **0** | 0 | **1** | 0 | ✓ | 1 | **1** | 1 |
| 1 | 0 | 1 | **1** | **1** | **0** | · | **0** | 0 | **1** | 1 | ✓ | 1 | **1** | 1 |
| 1 | 1 | 0 | **1** | **1** | **0** | · | **0** | 1 | **0** | 0 | · | 1 | **0** | 0 |
| 1 | 1 | 1 | **1** | **1** | **0** | · | **1** | 1 | **1** | 1 | ✓ | 1 | **1** | 1 |

* **Soundness**: $\Gamma \vdash \mathcal{A} \to \Gamma \vDash \mathcal{A}$ "Every provable statement is in fact true"
* **Completeness:** $\Gamma \vDash \mathcal{A} \to \Gamma \vdash \mathcal{A}$ "Every true statement has a proof"

## 4.3 Natural Deduction Rules

### Reiteration

| | | |
|---|---|---|
| $m$ | $\mathcal{A}$ | |
| $\therefore$ | $\mathcal{A}$ | R $m$ |

### Modus ponens

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \to \mathcal{B}$ | |
| $j$ | $\mathcal{A}$ | |
| $\therefore$ | $\mathcal{B}$ | MP $i, j$ |

### Modus tollens

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \to \mathcal{B}$ | |
| $j$ | $\neg\mathcal{B}$ | |
| $\therefore$ | $\neg\mathcal{A}$ | MT $i, j$ |

### Negation

| | | |
|---|---|---|
| $i$ | $\neg\mathcal{A}$ | |
| $j$ | $\mathcal{A}$ | |
| $\therefore$ | $\bot$ | $\neg$E $i, j$ |

| | | |
|---|---|---|
| $i$ | $\mid \mathcal{A}$ | |
| $j$ | $\mid \bot$ | |
| $\therefore$ | $\neg\mathcal{A}$ | $\neg$I $i$–$j$ |

### Indirect proof

| | | |
|---|---|---|
| $i$ | $\mid \neg\mathcal{A}$ | |
| $j$ | $\mid \bot$ | |
| $\therefore$ | $\mathcal{A}$ | IP $i$–$j$ |

### Double negation

| | | |
|---|---|---|
| $m$ | $\neg\neg\mathcal{A}$ | |
| $\therefore$ | $\mathcal{A}$ | $\neg\neg$E $m$ |

### Law of excluded middle

| | | |
|---|---|---|
| $i$ | $\mid \mathcal{A}$ | |
| $j$ | $\mid \mathcal{B}$ | |
| $k$ | $\mid \neg\mathcal{A}$ | |
| $l$ | $\mid \mathcal{B}$ | |
| $\therefore$ | $\mathcal{B}$ | LEM $i$–$j$, $k$–$l$ |

### Explosion

| | | |
|---|---|---|
| $m$ | $\bot$ | |
| $\therefore$ | $\mathcal{A}$ | X $m$ |

### Conjunction

| | | |
|---|---|---|
| $i$ | $\mathcal{A}$ | |
| $j$ | $\mathcal{B}$ | |
| $\therefore$ | $\mathcal{A} \wedge \mathcal{B}$ | $\wedge$I $i, j$ |

| | | |
|---|---|---|
| $m$ | $\mathcal{A} \wedge \mathcal{B}$ | |
| $\therefore$ | $\mathcal{A}$ | $\wedge$E $m$ |
| $\therefore$ | $\mathcal{B}$ | $\wedge$E $m$ |

### Disjunction

| | | |
|---|---|---|
| $m$ | $\mathcal{A}$ | |
| $\therefore$ | $\mathcal{A} \vee \mathcal{B}$ | $\vee$I $m$ |

| | | |
|---|---|---|
| $m$ | $\mathcal{A}$ | |
| $\therefore$ | $\mathcal{B} \vee \mathcal{A}$ | $\vee$I $m$ |

| | | |
|---|---|---|
| $m$ | $\mathcal{A} \vee \mathcal{B}$ | |
| $i$ | $\mid \mathcal{A}$ | |
| $j$ | $\mid \mathcal{C}$ | |
| $k$ | $\mid \mathcal{B}$ | |
| $l$ | $\mid \mathcal{C}$ | |
| $\therefore$ | $\mathcal{C}$ | $\vee$E $m$, $i$–$j$, $k$–$l$ |

### Disjunctive syllogism

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \vee \mathcal{B}$ | |
| $j$ | $\neg\mathcal{A}$ | |
| $\therefore$ | $\mathcal{B}$ | DS $i, j$ |

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \vee \mathcal{B}$ | |
| $j$ | $\neg\mathcal{B}$ | |
| $\therefore$ | $\mathcal{A}$ | DS $i, j$ |

### Hypothetical syllogism

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \to \mathcal{B}$ | |
| $j$ | $\mathcal{B} \to \mathcal{C}$ | |
| $\therefore$ | $\mathcal{A} \to \mathcal{C}$ | HS $i, j$ |

### Conditional

| | | |
|---|---|---|
| $i$ | $\mid \mathcal{A}$ | |
| $j$ | $\mid \mathcal{B}$ | |
| $\therefore$ | $\mathcal{A} \to \mathcal{B}$ | $\to$I $i$–$j$ |

### Contraposition

| | | |
|---|---|---|
| $m$ | $\mathcal{A} \to \mathcal{B}$ | |
| $\therefore$ | $\neg\mathcal{B} \to \neg\mathcal{A}$ | Contra $m$ |

### Biconditional

| | | |
|---|---|---|
| $i$ | $\mid \mathcal{A}$ | |
| $j$ | $\mid \mathcal{B}$ | |
| $k$ | $\mid \mathcal{B}$ | |
| $l$ | $\mid \mathcal{A}$ | |
| $\therefore$ | $\mathcal{A} \leftrightarrow \mathcal{B}$ | $\leftrightarrow$I $i$–$j$, $k$–$l$ |

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \leftrightarrow \mathcal{B}$ | |
| $j$ | $\mathcal{A}$ | |
| $\therefore$ | $\mathcal{B}$ | $\leftrightarrow$E $i, j$ |

| | | |
|---|---|---|
| $i$ | $\mathcal{A} \leftrightarrow \mathcal{B}$ | |
| $j$ | $\mathcal{B}$ | |
| $\therefore$ | $\mathcal{A}$ | $\leftrightarrow$E $i, j$ |

### De Morgan Rules

| | | |
|---|---|---|
| $m$ | $\neg(\mathcal{A} \vee \mathcal{B})$ | |
| $\therefore$ | $\neg\mathcal{A} \wedge \neg\mathcal{B}$ | DeM $m$ |

| | | |
|---|---|---|
| $m$ | $\neg\mathcal{A} \wedge \neg\mathcal{B}$ | |
| $\therefore$ | $\neg(\mathcal{A} \vee \mathcal{B})$ | DeM $m$ |

| | | |
|---|---|---|
| $m$ | $\neg(\mathcal{A} \wedge \mathcal{B})$ | |
| $\therefore$ | $\neg\mathcal{A} \vee \neg\mathcal{B}$ | DeM $m$ |

| | | |
|---|---|---|
| $m$ | $\neg\mathcal{A} \vee \neg\mathcal{B}$ | |
| $\therefore$ | $\neg(\mathcal{A} \wedge \mathcal{B})$ | DeM $m$ |

**Green:** basic rules.
**Orange:** derived rules.

More rules can be found in the "forall x: Calgary" book (p. 406).